The 8th IEEE International Conference on Intelligent Data Acquisition and Advanced Computing Systems: Technology and Applications
24-26 September 2015, Warsaw, Poland

An Image Threholding Approach Based on Cuckoo Search Algorithm and 2D Maximum Entropy

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***Abstract*—The image thresholding approach based on the basis of 2-D maximum entropy has better segmentation performance by the use of local space information of pixels, but it is unpractical for heavy computation required by this method. In the paper, an image segmentation technology based on cuckoo search and 2-D maximum entropy is presented, which views the seeking of 2-D maximum entropy of the image as a function optimization problem and uses the behavior of the obligate brood parasitism of some cuckoo species to simulate the process of searching optimal threshold. Furthermore, a local search strategy is employed to improve the results in the cuckoo search algorithm. The experimental results proves that compared with 2-D maximum entropy thresholding optimized with genetic algorithm, differential evolution algorithm and particle swarm optimization algorithm, the proposed method is able to get the optimal thresholds quickly and with a higher probability to get optimal threshold, which is a fast and robust image segmentation method.**

Keywords—Cuckoo Search Algorithm; 2-D Maximum Entropy; Thresholding; Image Segmentation

I. INTRODUCTION

Image segmentation is the basic task of digital image processing [1]. It divides an image into several meaningful parts by certain features of image to extract the region of interest from complex scenes [1-2]. Thresholding is a simple and efficient segmentation technology and it is assumed that one or several gray scale value could separate the target from the background [1-2]. The key to get ideal segmented results is to choose appropriate threshold. Many segmentation methods have been proposed, including thresholding method based on the minimum error, maximum entropy method, etc [3-4].[1] The common thresholding method based on 1-D histogram uses information of gray distribution of the whole image, which is implemented very fast. However, it ignore the local spatial information of the image. When an image contains noise or has a small target, the segmented result would be poor. Thresholding methods based on 2-D histogram utilize the overall gray level information of image and the local space information of neighbor pixel, with certain anti-noise ability , which improve the quality of image segmentation. When thresholding value is extended from 1-D to 2-D, the computational complexity would increase by hundreds of times compared with the 1-D thresholding method. In order to improve the performance of 2-D thresholding method, some optimization algorithms have been employed to search for the threshold of 2-D maximum entropy, which is widely concerned in recent years. For instance, J. Wang adopted genetic algorithm (GA) to optimize 2-D maximum entropy threshold method for getting the threshold value fast [5]. The particle swarm optimization algorithm (PSO) was also used to optimize 2-D Otsu and 2-D maximum entropy threshold [6-10]. Although the methods above have improved the efficiency of seeking the 2-D maximum entropy method, PSO and GA are easy to fall into the local optimum, and can't guarantee to obtain the optimal segmentation threshold. Therefore, optimization problems of 2-D maximum entropy thresholding method of image still deserve further study.

CS (Cuckoo Search Algorithm) is a new kind of heuristic optimization algorithm proposed by X-S Yang, from the University of Cambridge. Related tests have demonstrated that CS can keep a good balance between local search strategy and efficient exploration in the whole search space. CS algorithm has a better ability of global optimization compared with GA, differential evolution algorithm (DE) and particle swarm optimization [11-12]. Therefore, a novel thresholding method based on 2-D histogram and CS is presented in the paper.

The rest sections are structured as below. Section 2 explains the fundamental of thresholding method based on 2-D maximum entropy. Sections 3 introduces cuckoo search algorithm. The program of applying CS to gain the optimal threshold values is illustrated in section 4. Experimental results are displayed in section 5. In final, section 6 draws a conclusion.

[1] This paper is supported by Natural Science Foundation of China (41301371), funded by State Key Laboratory of Geo-information Engineering, No. SKLGIE2014-M-3-3 and Funded by Beijing Key Laboratory of Urban Spatial Information Engineering, No. 2014207.

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II. 2-D MAXIMUM ENTROPY THRESHOLDING

Assuming that an image with the height of *M* pixels and width of *N* pixels, $f(x,y)$ denotes the gray-level of the pixel at coordinate (x,y), and the average gray value of *k*×*k* neighbors (usually *k* is equal to 3) of a pixel is defined as (1):

$$g(x,y)=\frac{1}{k^2}\sum_{m=-k/2}^{k/2}\sum_{n=-k/2}^{k/2} f(x+m,y+n) \qquad (1)$$

2-D entropy of an image is calculated as follow. First, get the probability P_{ij} which count the gray level of unit pixel at coordinate (x,y) and the mean gray level of its neighbors as (2):

$$P_{ij}=\frac{1}{M\times N}\{n_{ij}\,|\,f(x,y)=i,g(x,y)=j;i,j\in 0\sim L-1\} \qquad (2)$$

where P_{ij} denotes the 2-D histogram of the picture and n_{ij} denotes the number of a pixel whose gray level is i and local mean gray level of 3*3 neighbors is j. L denotes the max value of gray level of the image. Fig. 1 shows the 2-D histogram of the image that is divided into four parts A,B,C and D by the threshold pair(*s,t*).

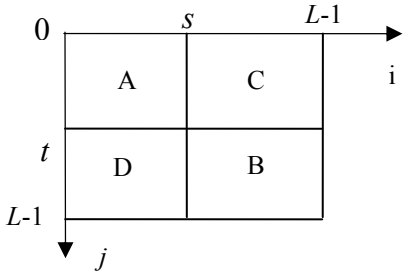


Figure 1. The 2-D histogram and the divided subarea.

Generally, A and B represent the background and the target area. Part C and part D consists of information mostly about the edges and noises, which usually can be ignored. In order to gain the more accurate segmented result, the information of the part C and the part D are saved.

The 2-D entropy of the part A can be calculated by (3):

$$H(A)=-\sum_{i=0}^{s-1}\sum_{j=0}^{t-1}\frac{P_{ij}}{P_A}\lg\frac{P_{ij}}{P_A}=\lg P_A+\frac{H_A}{P_A} \qquad (3)$$

Where $P_A=\sum_{i=0}^{s-1}\sum_{j=0}^{t-1}P_{ij}$ $H_A=-\sum_{i=0}^{s-1}\sum_{j=0}^{t-1}P_{ij}\lg P_{ij}$

Similarly, the 2-D entropy of the part B, part C and part D can be computed as below.

$$H(B)=\lg P_B+\frac{H_B}{P_B},\ H(C)=\lg P_C+\frac{H_C}{P_C},$$

$$H(D)=\lg P_D+\frac{H_D}{P_D}$$

The total 2-D entropy of an image is calculated by (4):

$$\psi(s,t)=H(A)+H(B)+H(C)+H(D) \qquad (4)$$

The optimal 2-D threshold pair (s, t) can be found by making the follow formula get the maximal value.

$$\psi(S,T)=\max(\psi(s,t)),\ 0\le s,t\le L-1 \qquad (5)$$

III. OVERVIEW OF CUCKOO SEARCH ALGORITHM

The cuckoo search algorithm is a newly proposed meta-heuristic algorithm, which mainly inspired by the conductions of cuckoo such as breeding by brood parasitism and searching by levy flights [11-12]. Levy Flights is a kind of random walk with alternate short distance exploring and long distance stochastic walking. By the means of levy flights, the searching range of solution is enlarged, so as to augment the chance of jumping out of local optimum. The basic rule of the brood parasitism about cuckoos can be summarized as follow:

(1) One cuckoo lays only one egg once and selects a nest at random to lay up the egg;

(2) In all of the randomly selected brood parasitism, the one in best position will be kept to the next generation;

(3) The number of the brood parasitism that can be used for laying egg is fixed. The probability that the egg can be discovered by the host bird is p_a.

If the egg has been distinguished as exotic egg, the host bird could choose to give up the egg or give up the nest and nest a new one. Simply, the last rule can be approximate to the probability of p_a that the nest can be replaced by the new brood (new solution). To the optimization problem, the constraint conditions are assumed as follows. (1) One brood can be put only one egg. And each egg represents one of the solutions about the problem. (2) Each egg of cuckoo represents a new solution. The new solution may be a better solution and can replace the old solution. According to the rule of the CS, one nest can be put only one egg, the position of the egg is decided by the position of brood. Therefore the solution of CS is indicated by the position vector of the nest. Based on the rule and hypothesis as above, the position of cuckoo is updated by (6).

$$x_i(t+1)=x_i(t)+\alpha\oplus Levy(\lambda),\qquad i=1,2,...,n \qquad (6)$$

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where $x_i(t+1)$ is the position of *i*-th cuckoo in generation *t+1* and $x_i(t)$ is the position in generation *t*. $\oplus$ denotes dot product, and α is the controlled parameter of step length, usually it is set to 1. Actually, (6) is a equation of random walk which is a Markov chain whose next location only relies on the current location and the transition probability. The Levy Flight provides the random search path which obeys the Levy distribution as (7).

$$Levy \sim u = t^{-\lambda} \qquad (1 < \lambda \le 3) \qquad (7)$$

Let $f(X)$ denote the goal function of the problem to be solved. The dimension of the solution space is *d*, namely $X = \{x_1, x_2, ...x_d\}$. The basic process of CS shows as follow [10-11].

Step 1: set parameters, and initialize a group of cuckoo, produce *n* nests with the initial position $X_i = \{x_{11}, x_{12}, ...x_{1d}\}$ randomly, where $1 \le i \le n$.

Step 2: get fitness value of each nest by fitness function and compare positions of all nests. Then record the fresh best position of all nests $p_g = (p_{g1}, p_{g2}, ...p_{gd})$.

Step 3: reserve the best position of the last generation and apply (6) to update the position of the nest.

Step 4: compare the new position of the nest to the last best position and keep the better one as the current best nest.

Step 5: produce a random number $r \in [0,1]$ and then compare it with p_a. if $r > p_a$, produce a new nest to replace the old one.

Step 6: judge if the end condition is met and if it is, go to step7, or return to step 2.

Step 7: output the best position of the all nest as the solution.

IV. Thresholding Approach Based on CS and 2-D Maximum Entropy

Regarding the 2-D maximum entropy of the image as a problem of function optimization, it can be handled by simulating the behavior of the obligate brood parasitism of cuckoos.

The main process of the proposed method is as below.

1) Determine the coding mode of the solution. Usually, an image has 256 gray scale levels. So the range of threshold *s* is [0,255] and for two-dimension thresholds, the range of the second threshold *t* is also [0,255]. Thus the real number pair(*s,t)*could be viewed as a threshold pair. It is corresponding to the position of the nest of cuckoos, where *s* denotes the first dimension threshold and *t* denotes the second dimension threshold.

2) Design the fitness evaluation function. By the virtue of 2-D maximum entropy based thresholding, (5) is adopted as the fitness function.

3) The detailed process of the algorithm of the proposed method is the same as the basic steps of CS in section 2.

4) In order to ensure that the solution is the optimum threshold, a local search strategy is used to make the result more precise [6]. A fluctuant threshold *B* is added to threshold pair (*s, t*) obtained by CS. Then the local exhaustive search will be used in the range from [*s-B, t-B*] to [*s+B, t+B*]. And the final threshold is (s*,t*). The whole procedure of the proposed method is shown in Fig. 2.

V. Experiment and Discussion

In order to test the performance of proposed method, this paper respectively chooses thresholding methods based on 2-D Fisher criterion, 2-D Otsu, GA, DE and PSO to combine 2-D maximum entropy method for a contrasting experiments. Here only list the results of three thresholding approaches on four images which respectively adopts the 2-D Fisher criterion thresholding method, the 2-D Otsu thresholding method and the proposed method. These testing images are remote sensing and infrared images. Fig. 3 shows the 2-D histograms of testing images.

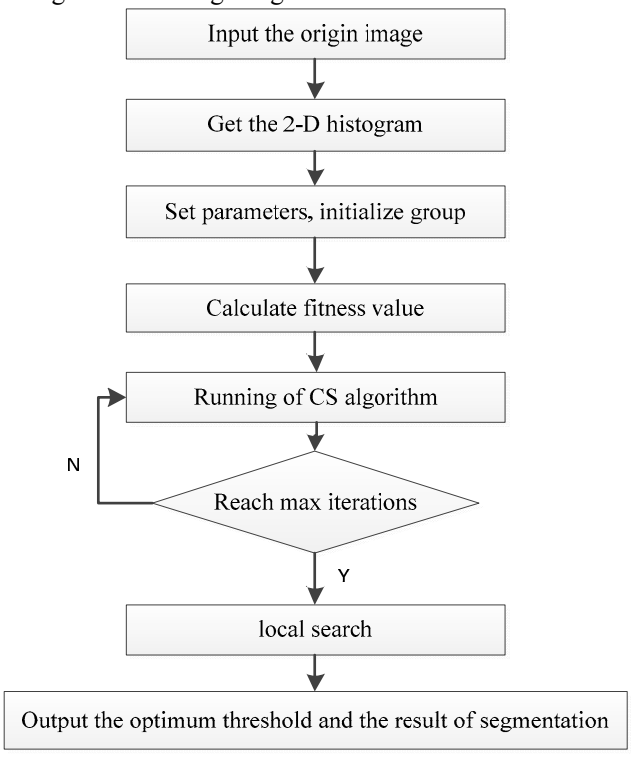


Figure 2. The process of the proposed methodology.

The experimental environment adopts Windows XP operating system, the 2.53GHz CPU, 2G RAM memory, and all algorithms are programmed and implemented in the Matlab 2012a. The main parameters used for these approaches are as follows: Fluctuation threshold B=10; the population size of all heuristic algorithms are 50, the

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maximum iteration $Iter = 30$. For GA, the selection rate is $ps = 0.9$; the crossover rate is $pc = 0.85$; the mutation rate is $pm = 0.05$. For PSO, the cognitive coefficient is $c_1 = c_2 = 2$, the maximum speed of particles is $v_{\max} = 200$; for DE, the scale factor is $f_m = 0.54$, crossover rate is $C_R = 0.7$ For CS, the control parameters is P_a=0.7. Considering that the GA, PSO, DE and CS are random probability algorithm in essence, the ability of optimization of all algorithms should be compared in a fair condition. For each tested image, all algorithms are run independently 50 times. The average fitness value and the number of times to get the optimal threshold with local search strategy are respectively listed out in Table I and Table II. The data in Table I is the average fitness value in condition that all algorithms don't implement local search. The data in the Table II represents the number of times to obtain the optimal threshold values when all algorithms make local search.

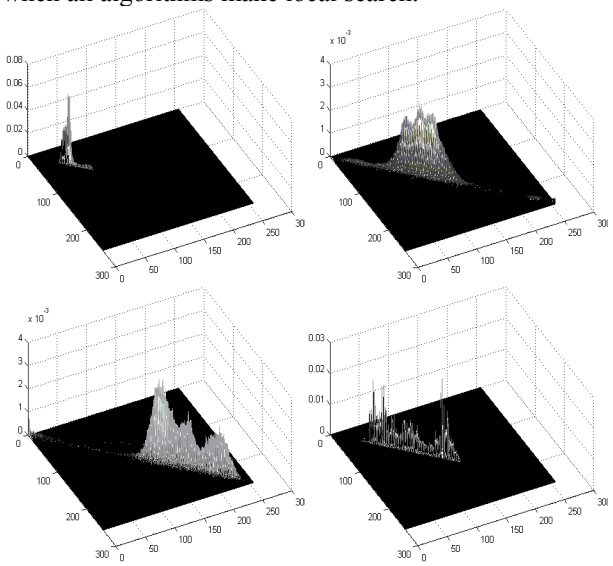


Figure 3. (a) The 2-D histogram of image I (b) The 2-D histogram of image II (c) The 2-D histogram of image III (d) The 2-D histogram of image IV.

Figures 4-7 respectively show the segmented results for 4 testing image by using different thresholding methods. The result of segmentation indicate that the quality of segmentation of 2-D Otsu is not so good, and the segmentation method based on CS algorithm with 2-D maximum entropy generally has better segmentation than the method based on 2-D Otsu and 2-D Fisher criterion for tested images. Especially when the target only takes up a small proportion of the whole image and there is only a small target, the proposed method would be more accurate.

The experimental data of GA, DE, PSO and CS algorithm to optimize 2-D maximum entropy method are recorded in Tables I-III. From the Table I, we know that the CS has the greatest average fitness value in all algorithms, which demonstrates that CS algorithm has the best optimal performance compared with the rest algorithms and the GA has the worst average fitness value in all algorithms. In a short, the analysis of effect of segmenting of different algorithms above is uniformed with analysis of the data in Table I. Take a further analysis of the data of the tested image I, II and IV in Table II. PSO and GA could get the optimal threshold with the probability less than 80%. Especially, GA only could get the optimal solution at 40% probability. DE algorithm has the average probability of 83% to obtain the optimal threshold. Considering the result that for image I2, CS obtained the optimal solution 49 times in 50 times operations. For 4 testing images, CS algorithm can obtain the optimal thresholds with the average probability of 93.5%, which show that compared with the other three algorithms, CS algorithms has the best performance.

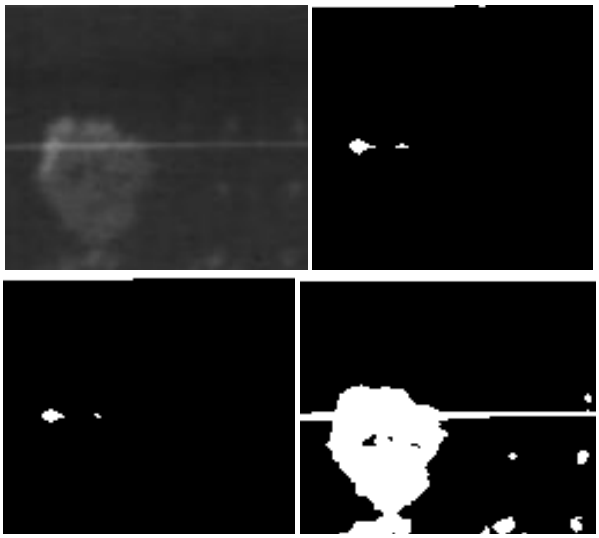


Figure 4. (a) The original image, (b) 2-D Otsu thresholding (c) 2-D Fisher criterion thresholding, (d) The proposed method.

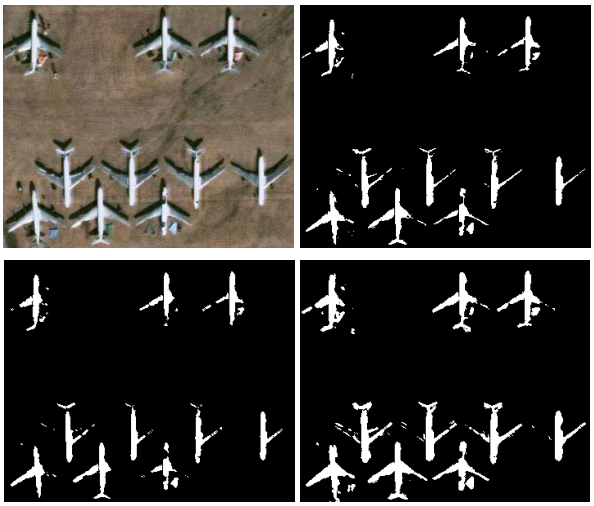


Figure 5. (a) The original image, 2-D Otsu thresholding (c) 2-D Fisher criterion thresholding, (d) The proposed method.

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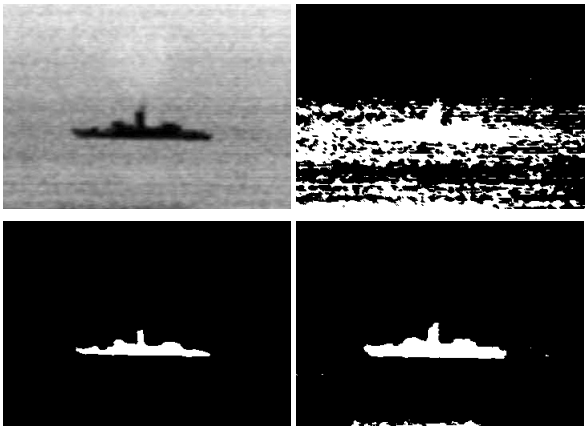


Figure 6. (a)The original image, (b) 2-D Otsu thresholding (c) 2-D Fisher criterion thresholding, (d) The proposed method.

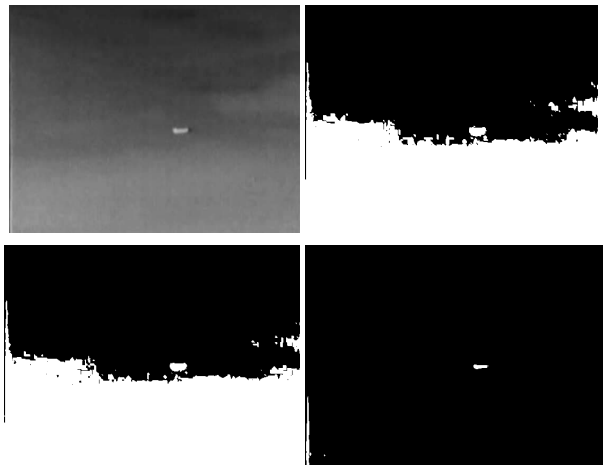


Figure 7. (a) The original image, (b) 2-D Otsu thresholding (c) 2-D Fisher criterion thresholding, (d) The proposed method.

Further, by making an analysis of computational efficiency, time consuming of thresholding method based on 2-D maximum entropy mainly lies in calculation of evaluation function. Exhaustive method needs to be done 256 * 256 = 65536 times operation for calculation on evaluation function. The proposed method in this paper only needs 1950 times objective function calculation, that is 50 (initial population) + 30*50 (iteration) + 20*20 (volatility threshold) = 1950 times. Approximately the required calculation of the proposed method is only one thirty (1950/65536 ≈ 1/30) of the exhaustive method. The actual operation time of the four algorithms is given in Table III, which is very close to theoretical time. In addition, Table III shows that the running time of GA is longer than other algorithms, and the performance of GA is the worst. The performance of PSO and of DE algorithm is close to CS. CS is the best among 4 algorithms tested in the paper. The average segmentation time of 4 testing image in CS algorithm is about 5 seconds, which could meet the requirements of some segmentation. In sum, the image threholding approach based on CS algorithm and 2D maximum entropy proposed in this paper has a better performance and the run time is not the shortest but the second shortest.

TABLE I. THE FITNESS VALUES OF DIFFERENT ALGORITHMS

Image	GA	PSO	DE	CS
I1	15.5367	15.5757	15.5784	15.5865
I2	24.6827	24.7354	24.7384	24.7392
I3	21.1396	21.1687	21.1765	21.1789
I4	16.3631	16.3755	16.4078	16.4427

TABLE II. THE NUMBER OF OPTIMAL THRESHOLD OF DIFFERENT ALGORITHM

Image	GA	PSO	DE	CS
I1	15	33	40	46
I2	35	47	48	49
I3	20	32	37	47
I4	28	38	41	45

TABLE III. THE REQUIRED RUN TIME OF DIFFERENT ALGORITHMS

Image	exhaustion	GA	PSO	DE	CS
I1	169.3421	5.932	5.734	5.670	5.720
I2	135.5572	4.116	4.071	3.988	4.050
I3	130.6637	5.765	5.338	5.299	5.354
I4	182.1790	6.300	5.190	5.107	5.123

VI. CONCLUSIONS

In the paper, an image segmentation technology based on cuckoo search and 2-D maximum entropy is presented. In order to test its performance, the proposed method are compared with the other methods, like 2-D maximum entropy thresholding method optimized with genetic algorithm, with differential evolution algorithm, with particle swarm optimization algorithm. The result of experiment show the proposed method has better performance than 2D maximum entropy methods respectively based on GA, PSO, DE. In general, the proposed method is an effective and stable approach for image thresholding segmentation. Our future work will be focused on multilevel thresholding of 2-D maximum entropy and improving CS algorithm for a better performance for image segmentation.

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